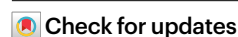


Expanded modelling scenarios to understand the role of offshore wind in decarbonizing the United States

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An assessment of decarbonization pathways in energy models reveals fundamental limitations in representing factors that are relevant for practical decision-making. Although these modelling limitations are widely acknowledged, their impact on the deployment of individual power generation types is not well understood. As a result, the societal value from such generation types could be vastly misrepresented. Here we explore a wide spectrum of factors that impact offshore wind deployment in the United States using a detailed capacity expansion model. Many factors prescribe a large future role for offshore wind, yet this diverges from what models often show. We extend the typically narrow modelling context through high spatial resolution, several cost and transmission possibilities and various energy-sector policies. We estimate offshore wind to constitute 1–8% (31–256 gigawatts) of total US generation by 2050. This wide range suggests an uncertain but potentially important regional role. Our expansive scenarios demonstrate how to address many limitations of decarbonization modelling.

The energy system is undergoing profound changes. Its transformation to support a decarbonized economy in many parts of the world is marked by multidimensional, complex and deeply uncertain processes¹. Energy models are used widely to understand the impacts from such changes to a power system and to inform decisions. As with any model, they represent an abstraction of a complex reality and are subject to inherent uncertainties. The limitations that result from these two defining features are commonly mitigated through explicit disclosure and by adopting a scenario approach². However, such an approach often falls short of illustrating the potential impact of alternative futures outside the study domain that is prescribed by data, computational, and conceptual limitations. We contend that this is a major issue for the analysis of deep decarbonization scenarios and relatively new technologies, for which limitations (and their stringency) come increasingly into focus. An improved understanding of the limited resolution, scope and variation in representing the siting, transmission, cost and policy characteristics of renewables can inform the growing

number and severity of trade-offs that arise with economy-wide decarbonization. For instance, a higher spatial resolution of geographic and human-use features is needed to identify feasible sites if renewables are to be deployed increasingly in more densely populated areas.

The principal limitations of energy modelling are well known^{3,4} and extend to the general analysis of complex systems^{5,6}. Time and space, uncertainty and transparency, growing complexities and the integration of human behaviour and social risks and opportunities are often identified as key modelling limitations³. We understand modelling limitations broadly as factors that reduce the suite of possible solutions ahead of a systems optimization. As such, modelling limitations are distinct from (mathematical) constraints to an optimization problem.

The limitations in modelling the energy system can arise from computational, resolution (spatial and temporal) or scoping boundaries⁷. Others highlight concerns about imposing constraints that make energy-model futures conform to an analyst's mental impression about how the future should unfold^{7,8}. Whereas modelling

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limitations are discussed extensively in principle, a systematic exploration of the different types that exist and an assessment of their impact on the deployed capacity of individual generation types is rare^{9,10}. If the importance of modelling limitations is unknown, decisions based on energy-system modelling could inappropriately assess and prioritize alternatives. This could yield a suboptimal power-system equilibrium and adversely impact policy targets, system reliability and costs for customers. A more thorough consideration of the spectrum of constraints and near-optimal solutions, including extreme events and low-probability, high-impact outcomes, has already been called for^{3,11–13}.

We reckon that such modelling limitations are particularly relevant for analysing the impact on the power sector from relatively new-generation technologies because their cost, interaction with transmission and policy, and siting tends to be highly uncertain. We consider offshore wind to be such a technology that is relatively new to the US power sector and is one that brings with it a suite of uncertainties. The offshore wind sector is at a nascent stage in the United States, with 42 MW of capacity installed. US state policies and federal targets aim for much greater future deployment levels¹⁴. Meanwhile, technology advancements, growing experience and standardization, and the upsizing of turbines and projects improve the economic case for deployment¹⁵. With an expansive resource, many stakeholders recognize the large potential role for offshore wind energy, yet the generation costs tend to be higher on average compared with land-based clean-energy alternatives, which suggests that growth is not assured. Moreover, the complexities regarding the variability of offshore wind and other variable renewable generation profiles, interactions with transmission, and strong regional differences in the resource make assessing offshore wind challenging. Assessments are further complicated by social acceptance for this new resource as well as the siting considerations related to the expansion of competing land-based resources. These factors, along with the intrinsic uncertainties in future market and policy conditions, necessitate both high-fidelity modelling and an assessment that covers a wider range of possibilities in the future of offshore wind. The limited deployment of offshore wind so far means that future uncertainties concerning the siting, permitting and other challenges are important.

Here we address these research gaps through a systematic assessment of the impacts from key modelling limitations on the deployment of offshore wind energy in the United States. Specifically, we explore the question of how sensitive offshore wind deployment is to factors that shape deep decarbonization pathways in the US power system, such as siting, transmission, policy and technology costs. To achieve this, we address several of the identified modelling limitations. We simultaneously introduce very high spatial and temporal resolution in combination with various uncertainties related to transmission, load growth, technology costs and energy-sector policies in a consistent scenario framework. Offshore wind energy is one of many technologies that could be used to assess modelling constraints. We focus on a single generation resource because impacts on its deployment using a scenario framework can be more clearly identified. The choice of offshore wind was further motivated because we consider its role in the United States to be generally understudied in power-systems analysis, particularly in comparison with its anticipated role by many stakeholders. On the role of offshore wind energy in the United States, the literature typically focuses on technology-specific analysis and does not examine system interactions^{16–19}. Some systems-level decarbonization studies have included offshore wind energy as a candidate technology, although this research rarely reports results, explores sensitivities or explicitly addresses the modelling limitations that are specific to offshore wind^{20–22}. Other studies use global integrated assessment models, which lack the temporal, spatial and technological detail needed to capture complex electric-sector planning decisions, including the higher fidelity required to address modelling limitations

that are common for offshore wind energy^{23,24}. Studies with greater detail on offshore wind tend to be focused on continental Europe (for example, refs. 25–29).

We advance the existing literature in several ways. Using a detailed capacity expansion model of the US electric sector, we investigate offshore wind deployment and its system interactions under deep decarbonization scenarios. Our scenario framework includes a wide spectrum of future conditions to explore the uncertainties related to energy policies, demand growth, technology costs, transmission expansion and siting regimes. This combination of scenarios is tailored to offshore wind, given the unique characteristics of the technology. We show that US offshore wind deployment beyond the current state targets can occur with stringent decarbonization efforts, possibly reaching 256 GW and 8% of total US generation by 2050. Even under deep decarbonization conditions, deployment is sensitive to other factors, such as the accessibility and deliverability of competing onshore clean-energy resources. At a regional level, offshore wind may play an even more important role in decarbonizing the economy, with up to 20% of total generation along the Atlantic coastal regions by 2050.

Factors shaping offshore wind deployment

We model 19 scenarios designed to span a broad spectrum of future conditions that could impact the long-term role of US offshore wind energy. The scenarios are grouped into four categories: (1) policy and demand, (2) technology cost and availability, (3) transmission and (4) siting constraints for offshore wind and alternative generation options. The first group includes scenarios with clean-energy policies from 2021 only and extends to scenarios that fully eliminate carbon dioxide (CO₂) emissions in the power sector, and for a range of demand-growth projections. The remaining scenario groups are centred on a single emissions-reduction and demand-growth outcome, consistent with a deep decarbonization trajectory, to isolate variations in other drivers for the deployment of offshore wind (see Methods and Supplementary Tables 1, 2, 3 and 4 for the detailed assumptions).

To assess the impact of these limitations, we use the regional energy deployment system (ReEDS) capacity expansion model and develop scenarios of the US electricity sector up to 2050. The ReEDS model is designed with high spatio-temporal resolution to capture the unique characteristics of renewable energy and competing clean-energy options, along with policy, transmission, resource and other grid constraints. Importantly, the high resolution of the system used is necessary for modelling some of the important dimensions, namely transmission and siting.

Clean-energy policies

Analyses of the impact from broader energy-sector policies on offshore wind deployment are often limited. We model a range of scenarios with varying emissions targets in the power sector, including existing policies only (as of June 2021) and national CO₂ reductions of 75%, 85% and 95% in 2050 (from 2005 levels). These scenarios are then overlaid with various projections for electrification in the future energy system that yield changes to the load growth (from 1.0% per year to 3.4% per year) and demand profiles (for example, increased winter peaks)^{30,31}. In many of these scenarios we find limited offshore wind deployment beyond that which states have already committed to today. We estimate approximately 30–40 GW of offshore wind capacity by 2035. If power-sector emissions decline by more than around 80% below the 2005 levels and are combined with widespread electrification through the use of electric vehicles, heat pumps and other end-use electric technologies, offshore wind deployment can expand beyond these levels (Fig. 1). Under our ‘core’ scenario, with an emissions cap of 80% below the 2005 levels by 2035 (95% by 2050) and electrification-driven demand growth resulting in 6,710 TWh of demand by 2050 (compared

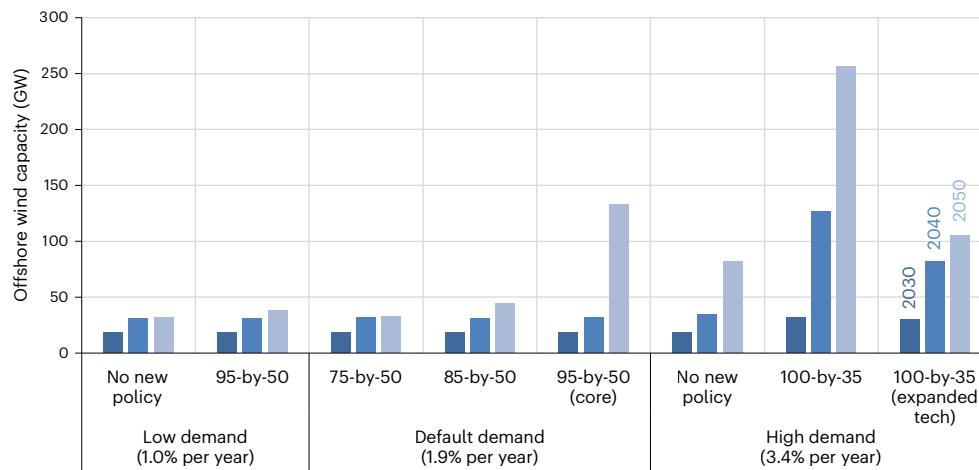


Fig. 1 | Offshore wind capacity in the policy and demand scenarios.

The x-axis labels refer to the national carbon policy implemented (first level) and the electricity-demand assumptions implemented (second level, which also includes the average annual load growth). The combination of carbon policy and demand defines the scenario. ‘No new policy’ indicates no national carbon policy and includes only existing (as of June 2021) state-level policies, to result

in national-power-sector carbon emissions that are 59% below the 2005 levels by 2050 (44% by 2035). ‘95-by-50’ refers to a national carbon-emissions constraint that lowers US power-sector emissions to 95% below the 2005 levels by 2050; ‘75-by-50’ and ‘85-by-50’ are defined similarly. See Methods and Supplementary Table 1 for more details.

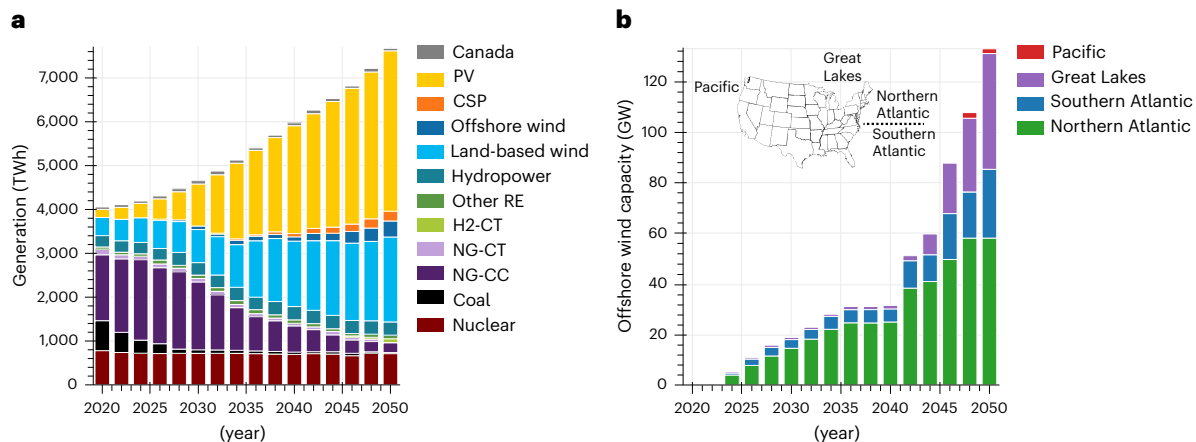


Fig. 2 | Electricity generation and offshore wind capacity in our core scenario. a,b, Electricity generation (a) and offshore capacity by region (b) in the core scenario. RE, renewable energy; H2-CT, hydrogen combustion turbine; NG-CT, natural gas combustion turbine; NG-CC, natural gas combined cycle. ‘Canada’ denotes net energy imports from Canada.

with 3,690 TWh in 2020), the offshore wind capacity reaches 133 GW by 2050. Development starts in the northern Atlantic region and expands substantially during the 2040s to the southern Atlantic and Great Lakes regions (Fig. 2).

A limit on emissions in combination with energy-sector electrification increases the overall demand for clean electricity, including offshore wind energy. In fact, the scenario with the greatest amount of offshore deployment (256 GW, 8% of 2050 generation) is one that achieves 100% carbon-free electricity and very high electrification (that is, the ‘admin target’ scenario). Several secondary effects can impact the pattern of offshore wind deployment. The additional demand for clean electricity increasingly exhausts the highest-quality sites for competing land-based wind and solar photovoltaics (PV). Electrification could also lead to demand profiles that are more favourable for offshore wind deployment by shifting the peak-demand periods to the winter season, which is more closely aligned with typical offshore wind generation profiles³². By extending energy modelling to an assessment of these policies and load growth, we obtain a more nuanced understanding that they only matter for offshore wind deployment when the policies become relatively stringent.

Technology alternatives and costs

Although technology cost sensitivities have been commonly explored in scenario studies^{20,21}, the availability of generation technologies that are emerging for first-time commercial-scale deployment have been less frequently studied. For an evaluation of ‘emerging’ technologies with a limited deployment record, cost and availability sensitivities are particularly insightful because these technologies are often on the margins of deployment. The emerging status of these generation technologies means that there is a fair degree of uncertainty as to when they will become available for commercial deployment and at what cost of generation. See Methods for the assumptions around technology costs as well as the availability of new technologies.

Across all of our studied scenarios, new generation capacity comprises predominantly PV and land-based wind. Under the core scenario, electricity from land-based wind and PV combined make up 73% of the total 2050 generation (1,800 GW from PV and 556 GW from land-based wind). The combined amount of land-based wind and PV is relatively consistent (68–78% by 2050) across scenarios with a 95% emissions reduction by 2050, although the mix between land-based wind and PV varies depending on their relative costs and siting constraints

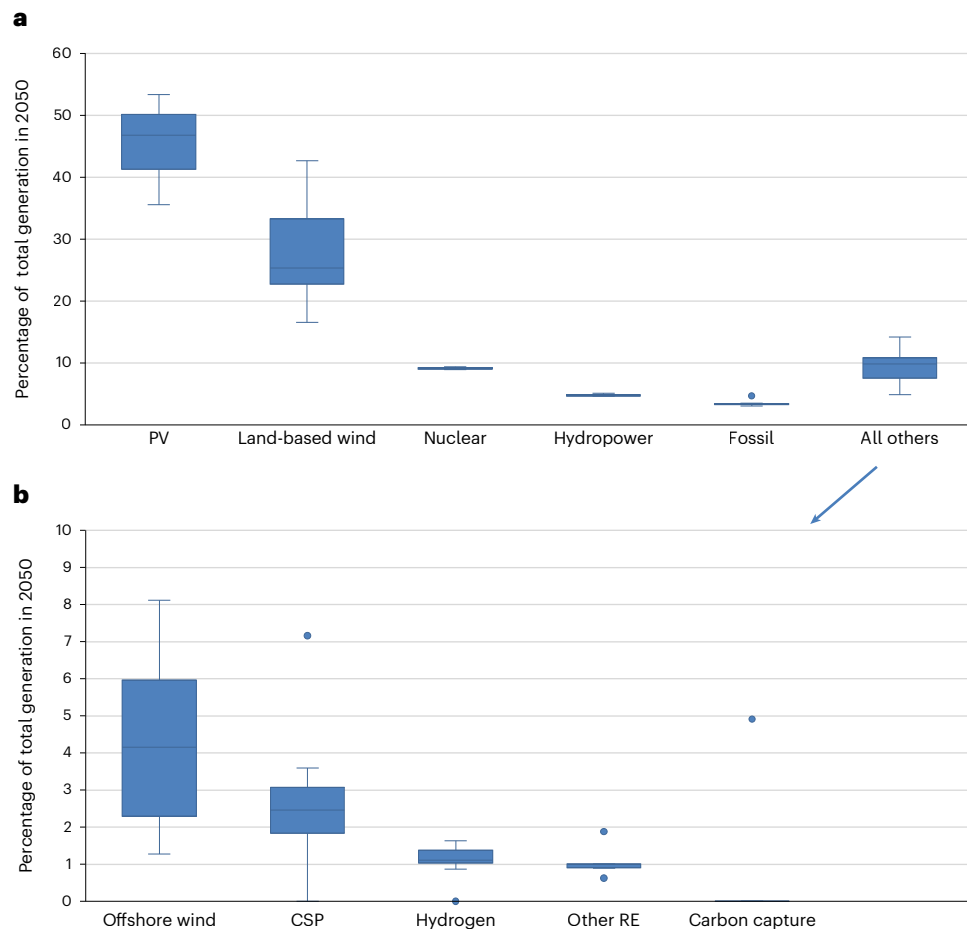


Fig. 3 | Generation fraction in 2050 for the default-demand 95% CO₂ reduction scenarios. a,b, Percentage generation in 2050 by technology in the 12 scenarios with the default electricity demand ('EFS High' (see 'Data' in the Methods)) and 95% CO₂ emissions reductions by 2050, where the 'all others' category in **a** is the sum of the technologies shown in **b**. The 'fossil' category includes NG-CT, NG-CC and coal technologies (all without CCS); 'carbon capture' includes biopower with CCS and natural gas with CCS; 'hydropower' includes net imports from Canada. Each box and whisker chart is based on data from the 12 scenarios (that is, sample

size = 12). Boxes are divided into quartiles using the exclusive median method, with a box drawn between the first and third quartiles and the centre line marking the median. Whisker lengths are based on 1.5 times the interquartile range (IQR), that is, the top whisker extends from the top of the box to the largest value such that the whisker length is less than or equal to 1.5 times the IQR, and the bottom whisker extends from the bottom of the box to the smallest value such that the whisker length is less than or equal to 1.5 times the IQR. Values outside of the whiskers are shown as dots.

(see Supplementary Fig. 5). With the same emissions policy, nuclear, hydropower and fossil generation are also consistent between scenarios. Fossil generation is limited by the emission cap. When carbon capture and sequestration (CCS) is available, the amount of fossil generation increases (Supplementary Fig. 5). Very little new hydropower and nuclear capacity is installed. We assume that no retirements for hydropower take place. Nuclear plant retirements are determined by either their announced retirement date or the assumed maximum lifetime of 80 years. However, other technologies that make up the remaining generation vary considerably among the scenarios (Fig. 3). Our scenario framework reflects several uncertainties for the emerging technologies (Fig. 4). For example, a scenario without new concentrating solar power (CSP) and geothermal contributors (that is, the 'limited tech' scenario) can lead to 24 GW of offshore wind capacity by 2050 in addition to the core scenario (133 GW total); conversely, when CCS and advanced nuclear are assumed (that is, in the 'expanded tech' scenario), offshore wind deployment is reduced by 70 GW compared with the core scenario. These impacts can be even more pronounced from a regional perspective. For instance, the limited-tech scenario leads to a greater deployment of floating offshore wind in California (see Supplementary Information for regional details and an additional sensitivity analysis for California). The extent of the impact from technology costs and availability depends on how they compare relatively

among the technologies. Lower offshore wind costs, while holding all else equal (that is the low offshore wind cost scenario), result in offshore wind deployment well above the core scenario (211 GW versus 133 GW by 2050).

Transmission expansion

The transmission trajectory is a key determinant of future deployment, especially for technologies with a high degree of spatially varying resource quality and potential. An often-cited benefit of offshore wind energy is its proximity to coastal load centres^{33,34}, whereas the best land-based wind and PV sites are often remote from such regions. Thus, the economically advantageous choice between offshore wind and these other resources is sensitive to the transmission trajectories. The future cost of transmission expansion, along with the competing use considerations and social acceptance, introduces considerable uncertainties. Modelling a transmission scenario requires adequate spatial resolution, which we detail in the Methods and Supplementary Table 3.

Under the core scenario, we allow transmission expansion within only 12 transmission planning regions. Limiting transmission expansion to these planning regions results in a concentration of offshore wind deployment in northeastern regions of the United States. Offshore wind makes up more than 20% of the 2050 generation in New York

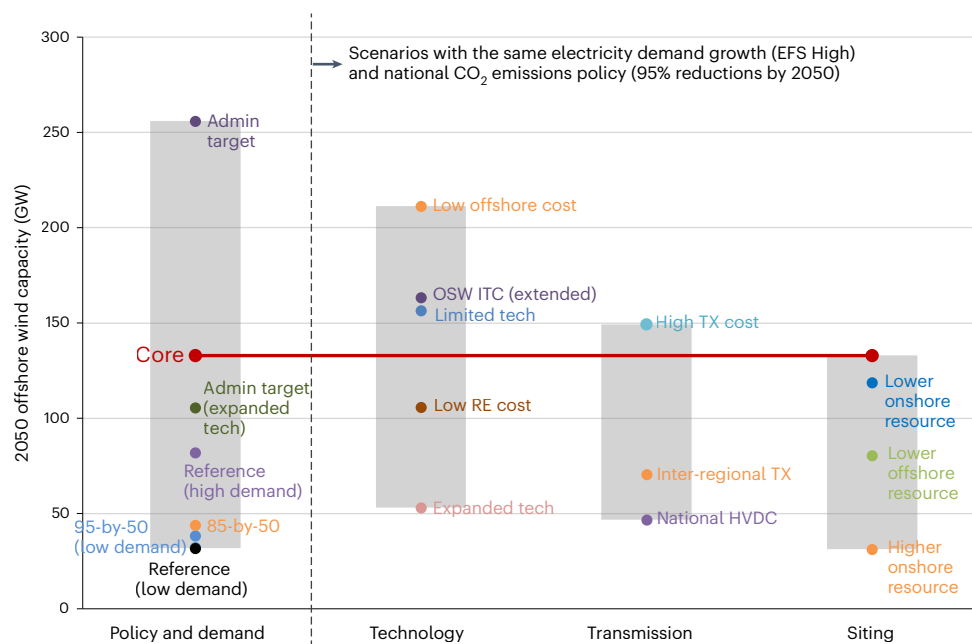


Fig. 4 | Offshore wind capacity in 2050 across all scenarios. The offshore wind capacity in 2050 across all scenarios is separated by scenario group. The grey shaded areas span between the maximum and minimum values in each scenario group. Colours have been added to differentiate the dots. Note that the ‘75-by-50’

scenario is not shown because it overlaps the ‘Reference’ scenario dot. OSW ITC, offshore wind investment tax credit; TX, transmission; HVDC, high-voltage direct current.

and New England, whereas offshore wind generation is 5% nationally. Under a scenario with such constraints on transmission expansion relaxed (that is, the ‘inter-regional transmission’ scenario), the additional competition from interior resources—even taking into account the costs of transmission—overcomes these offshore wind advantages and reduces the 2050 offshore wind capacity to 71 GW from 133 GW in the core scenario. Allowing even more flexibility in transmission expansion, and including a high-voltage direct-current macrogrid that offers a lower-cost option for moving power across long distances, further reduces the estimated offshore wind deployment by 2050 (47 GW in the national ‘high-voltage direct-current’ scenario). Conversely, if transmission is both constrained to within planning regions and more costly (that is, the ‘high transmission cost’ scenario), offshore wind deployment increases to 16 GW above the core scenario by 2050. This scenario reflects the combination of limited coordination for transmission planning between regions and higher installation costs for underground transmission or complex routing to overcome the challenges of transmission siting.

Siting of renewables

Analysing the siting of renewables in long-term planning models requires sufficient spatial fidelity since the impacts of siting are not uniform across different regions. For example, in more densely populated regions, the siting of wind or PV projects tends to face a greater level of resistance than in areas that are more sparsely populated. Scenario studies seldom examine the impacts of siting on future generation mixes and costs. We explore how different siting restrictions can impact the deployment of offshore wind. Our core scenario represents a siting regime where land-based resources are constrained by regulatory restrictions on protected lands, state and county ordinances as well as several factors related to competing human use. For land-based wind, these constraints include a setback requirement of threefold the maximum tip height from all roads, buildings and other infrastructure, exclusions from shadow flicker and other regulatory or physical challenges to the siting of wind turbines^{36,37}. As a result, the land-based

wind-resource potential is only 2.1 TW for the contiguous United States as a whole, but with a concentration of excluded land areas in more densely populated regions along the coasts and Midwestern states. Under a scenario that uses less restrictive siting assumptions (for example, a setback requirement of 1.1-fold), the land-based wind-resource potential is 3.2 times greater (the ‘higher onshore resource’ scenario). The greater land-based wind-resource potential of this scenario relative to the core situation comes predominantly from sites closest to load centres. With greater land-based wind resource available for deployment, offshore wind deployment is estimated to be much lower (31 GW in 2050) (Fig. 5). The developable offshore wind resource and, ultimately, its offshore deployment can also be impacted by alternative uses, such as shipping, marine wildlife and views considerations³⁸. In a scenario with more restrictive siting for offshore wind (the ‘low offshore resource’ scenario), we find a strong reduction in the estimated offshore wind capacity by 2050 (81 GW) compared with the core scenario. System costs are another measure that reveals the relevance of siting restrictions: the more restricted they are, the higher the total electric-system costs, which vary by nearly US\$308 billion (10% of the present value 2022–2050 costs for the core scenario) across the three siting scenarios described here—a similar magnitude of impact as the technology cost and availability scenarios (see Supplementary Figs. 9 and 10). Under the core scenario, capital investments into offshore wind energy total US\$95 billion or less than 3% of the present value costs of the total system.

Although we report modelling results at the regional level, our analysis is designed as a national-scale assessment using wide-ranging scenarios. Such an analysis can also be applied to a state or region. To demonstrate this, we model the sensitivity from our core scenario that includes California’s offshore wind target of 25 GW by 2045, which is currently a ‘preliminary planning goal’ that establishes a ‘potentially achievable but aspirational’ target³⁹. In this scenario, offshore wind comprises 15% of California’s generation in 2050 and displaces CSP and solar PV, where the system costs are nearly identical to those of the core scenario (see Supplementary Note 1).

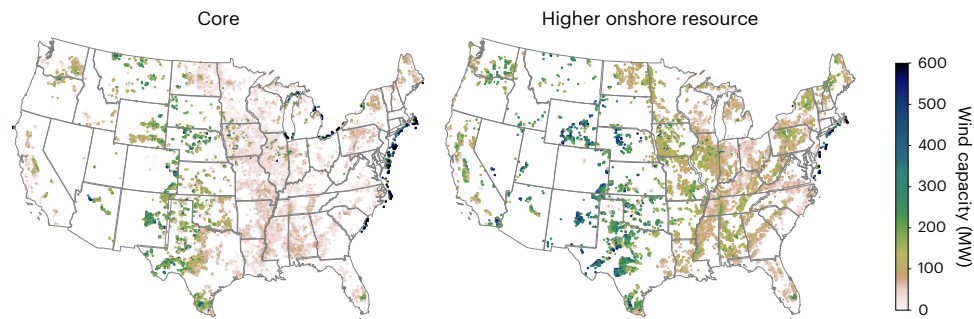


Fig. 5 | Wind deployment in 2050 by site. Land-based and offshore wind deployment by site in 2050 for the core and higher onshore resource scenarios. Each deployed wind site is shown as a coloured marker on the map, with the colour corresponding to the deployed capacity at that site.

Discussion

Our results suggest an important role for offshore wind in the United States at a regional scale under deep decarbonization trajectories (Supplementary Fig. 5). Across our scenarios, up to around 20% of total generation in the Atlantic coastal regions will be served by offshore wind by 2050. In a sensitivity analysis, California offshore wind provides 15% of the state's 2050 electricity needs. Compared with today, these outcomes would represent a drastic increase in the offshore wind capacity. On a national scale, one may conclude a much more modest role for offshore wind (an average of 4% of total generation across 12 key scenarios) in comparison with land-based wind and PV resources (27% and 46%, respectively) and considering the offshore wind total resource potential of more than 4,300 GW (see Methods). Furthermore, across our scenarios, the range in offshore wind deployment of 1–8% of total generation (31–256 GW) by 2050 is considerable, and reflects the broader uncertainties regarding the future energy system and the great challenges in energy- and wind-sector planning. These findings could have far-reaching implications for power-sector planning and investments into the supporting infrastructure, such as manufacturing, transmission, ports and dedicated vessels. Currently, such supporting infrastructure is very limited in the United States, and its establishment may require the mobilization of US\$22 billion in capital investments as well as a workforce of more than 35,000 individuals for 30 GW of offshore wind capacity alone⁴⁰. The uncertainty in the magnitude and timing of offshore wind investments that can be inferred from our modelling results calls for greater coordination among local, state and federal planning authorities.

Our modelling of a larger number of scenarios indicates a broader range of possible offshore wind deployment compared with nearly all previous US decarbonization studies in the literature. Earlier US-focused analyses often do not distinguish between land-based wind and offshore wind, let alone conduct sensitivity analyses around factors that matter for the deployment of offshore wind. We expose key modelling limitations in existing research to a 'stress test' that combines multiple factors not often considered by traditional-scenario frameworks. Through our modelling advancements we find that offshore wind deployment in the United States is highly sensitive to key features of a decarbonized power system: the stringency of decarbonization policies, the generation cost differential between technologies, future transmission expansion and siting limitations of renewable-energy-generation assets. Our results illustrate how a single limitation or a combination of limitations can impact the deployment and can lead to starkly different conclusions. Omitting such factors from the analysis could mean that the potential of offshore wind energy to mitigate climate change is not appropriately captured. The greater role for offshore wind in US regions and scenarios that have siting limitations for land-based wind, transmission-expansion constraints and limited technological portfolios suggests that countries with such land use, regulatory, resource and public-acceptance constraints

(combined with decarbonization targets) are those where offshore wind deployment could be higher. European countries, Japan and Taiwan are examples where spatially detailed modelling is valuable for assessing the economic potential of offshore wind, and although some studies have begun to examine this role^{41–43} our modelling indicates the value in conducting a range of sensitivity analyses that focus on region-specific policy, market and technology drivers of offshore wind and its interactions with other electric-sector technologies.

Furthermore, the sensitivity of our results across and within the scenario groups highlights that all the factors considered can play an important role. Thus, an inability to represent these factors will prevent some of these important dimensions from being considered. In particular, our analysis highlights the importance of transmission and siting, which require high-spatial-resolution modelling to be captured and understood. A multi-zonal network model appears to be critical for us to understand how transmission constraints impact offshore wind deployment. Similarly, we model investment options for about 60,000 land-based and 8,000 offshore 'sites' and their corresponding resource potential (see Methods), which enables us to assess the local effects from siting constraints explicitly^{36,44}.

Even under conditions that are most favourable for offshore wind, we assert, however, that the deployed offshore wind capacity may fall short of stakeholders' expectations. A total of 17.6 GW by 2035 has been procured by utilities so far to meet state commitments that amount to a total of 39.3 GW of offshore wind by 2040. In February 2022, six new offshore wind lease areas were awarded to offshore wind developers for a total of US\$4.4 billion, attesting to the financial commitments and contributing to a project pipeline of 18.6 GW that is currently at various permitting stages⁴⁴. In several of our study scenarios, offshore wind deployment beyond that which has already been committed to by US states today does not occur, even by 2050, unless a deep decarbonization scenario with at least an 80% reduction in emissions and a high load growth is assumed (Fig. 1). Furthermore, the location of deployed sites varies from what is expected in practice. For instance, several planning activities are underway along the US Pacific Coast to issue offshore wind lease areas, such as the adoption in 2022 of an offshore wind goal of 2–5 GW (2030) and 25 GW (2045) by the California Energy Commission³⁹. However, our results suggest a much lower deployment in this state across our studied scenarios, with a total of only 1.8 GW on average in 2030 and beyond, apart from in the limited-tech scenario, which shows 14 GW by 2045 and a sensitivity that enforces the 25 GW target (see Supplementary Information).

Several reasons could explain these differences. First, decision-makers may already have internalized a deep decarbonization scenario and are planning accordingly. In a similar vein, there is a chance that decision-makers may prefer to procure energy from within the regions they have (some) jurisdiction over (such as offshore wind), rather than relying on imports from outside their jurisdictional boundaries (for example, solar PV or onshore wind) that are dependent

on inter-regional transmission being constructed. Second, there is the possibility that we have omitted key factors in our scenario analysis, such as widespread sector coupling, demand flexibility, impacts from climate change or capturing the dynamic relationships that are characteristic of the energy sector. As an example of this latter point, clean-energy policies could drive up the demand for renewable energy that in turn will lead to accelerated cost reductions through ‘learning by doing’. Our scenario framework examines these types of factors, but the assumptions are statically applied and the dynamic interactions between them are not captured. Third, more fundamental limitations in energy modelling may constrain the domain of possible solutions, such as the scope of the energy model itself or several near-optimal solutions for offshore wind deployment that are not identified by our modelling^{13,45}.

Conclusion

We draw three primary conclusions from our analysis. We find that US offshore wind deployment beyond current state targets occurs with stringent decarbonization efforts. At a regional level, offshore wind may play an even more important role by 2050, particularly along the Atlantic coastal regions. This could imply substantial investments into supporting infrastructure, workforce development and coordination needs among the planning authorities. However, even under scenarios with a concerted effort to reduce emissions in the US economy, we find that offshore wind deployment in the United States is highly sensitive to the stringency of decarbonization policies, the generation cost differential between technologies, the transmission topology and the siting limitations of renewable-energy-generation assets.

Our results also illustrate how a single limitation or a combination of limitations can impact deployment and lead to starkly different conclusions. Omitting such factors from the analysis, or via an inability to adequately represent such factors in a model, could mean that the potential of offshore wind energy to mitigate climate change is not optimally captured. Several of the scenario dimensions that we have investigated—namely around transmission and the local siting of renewable resources—require high-spatial-fidelity modelling for their proper exploration. Using a detailed model that is able to represent the location dependence and variability of offshore wind and competing renewable resources can identify important interactions between offshore wind deployment, transmission development and local siting factors. Such a type of analysis is particularly important for relatively new technologies because they may require a combination of these factors to see any modelled deployment at all (as we have seen here). We conclude that a wide scenario matrix, as well as a higher spatial and temporal resolution in the representation of key power-system features, can resolve many of the limitations we have identified in this study. However, the inherent limitations of modelling remain, and a careful interpretation and presentation of the results is advisable.

For future studies that extend beyond the context of our US modelling, we suggest several broader implications on the basis of our analysis: more meaningful results can be obtained through a more detailed investigation and resolution of the modelling constraints; a clearer articulation of the inherent modelling constraints can mitigate the risk of misleading policy conclusions being drawn; and the selection of a model that is tailored to the research question can produce more meaningful results. We suggest that a larger body of contextual studies will explore the impacts of addressing or mitigating the key limitations, similar to the type of complementary ‘off-model analyses’ that have been proposed by some to evaluate underappreciated extreme events¹¹.

Of the many limitations identified in the introductory section, we have focused on addressing spatial and temporal resolution in combination with key uncertainties that pertain to offshore wind technologies in the United States. Many other modelling limitations remain to be

explored (for example, the integration of human behaviour and social risk or scoping boundaries) from the perspective of other technologies and global power-sector systems.

Methods

Approach

Research was executed in three phases. First, we identified potential limitations in power-sector modelling by contrasting US offshore wind sector developments from ref. 14 with a detailed review of modelling assumptions in the capacity expansion model ReEDS. Second, we developed a set of scenarios in which each mitigates a modelling limitation, including a higher spatial resolution of renewables’ siting, a variety of transmission topologies and energy policies, to develop a suite of cost and technology availability scenarios. Third, we contrasted the results from the ReEDS scenarios with empirical evidence about the possible future development pathways of offshore wind energy.

Regional energy deployment system (ReEDS) model

The ReEDS model is a capacity expansion model for the electricity sector of the conterminous United States^{37,46}. It simulates changes to the power system—through new investments in generation, storage and transmission capacity as well as the retirement of existing infrastructure—up to 2050, in two-year periods. For each period, ReEDS identifies the least-cost portfolio given the input assumptions about future electricity demand, planning and operating reserve requirements, resource and system constraints, and technology cost and performance parameters. Technologies modelled in ReEDS include coal (with and without carbon capture), natural gas combined cycle systems (with and without carbon capture), natural gas combustion turbines, nuclear (conventional and small modular reactors), hydropower (various options), geothermal (hydrothermal and enhanced geothermal), biopower (conventional and with carbon capture), CSP (with and without thermal energy storage), PV (utility and distributed), land-based wind, offshore wind (fixed bottom and floating platform), battery storage (2–10 h duration), pumped storage hydropower, hydrogen combustion turbines, direct air capture and transmission (alternating-current (AC) and direct-current (DC) transmission). Hydrogen production is also modelled endogenously and includes two pathway options: electrolyzers and steam methane reforming with carbon capture.

ReEDS is designed with particular considerations for the unique aspects of renewable-energy technologies, including the variability and uncertainty of wind and solar generation and the regional characteristics of all renewable-energy resources. High spatial resolution in the model enables a detailed representation of the resource limits and development exclusions, transmission constraints and expansion, and state and regional policies. Although there are multiple spatial scales to the model, the primary grid network is represented by 134 model balancing areas in which supply and demand are balanced, electricity flows are modelled and most technologies are represented. Other larger regions comprising multiple balancing areas are used to represent political or market boundaries (for example, state boundaries to enforce state renewable portfolio standards and regional transmission organizations). In the version of ReEDS used for this analysis, wind energy is modelled with a much higher resolution. For land-based wind energy, there are about 60,000 potential sites represented in the model, each of which represents an 11.5 km grid cell. Each site is parameterized with a resource potential (measured in megawatts), annual and hourly capacity factors (for tractability, the hourly profiles are aggregated for each neighbouring (4 × 4) group of 16 sites), a grid connection cost and an economies-of-scale adder. Offshore wind energy is represented similarly and at a similar resolution (about 8,000 sites with a 10.6 km resolution). These developable-potential data and cost-and-performance data are from the renewable-energy-potential geospatial model^{36,38} that accounts for multiple land and water exclusions (see the ‘Data’ section below).

As with the spatial dimension, ReEDS includes multiple timescales depending on the relevant aspects considered⁴⁷. The core linear programming optimization in the model, which informs the investment decisions, disaggregates each year into 17 time slices (four diurnal periods for each season and a superpeak period); however, the model includes a separate module—which is called in between the solve years (for this analysis, we use two-year solve periods between 2020 and 2050)—that uses hourly data to estimate the parameters that are passed to the core optimization. Specifically, wind and solar curtailment, the effects that new storage and transmission will have for reducing curtailment, and the storage energy value are estimated using a chronological hourly economic dispatch model for one year. The capacity credit, the fraction of the nameplate capacity that will contribute to planning the reserve requirements of wind, solar and storage, is estimated by examining the generation profiles during seasonal net peak hours using seven years of hourly data.

The structure of the ReEDS model summarized above provides a detailed characterization of the wide-ranging options to meet grid and policy needs. Ultimately, these characterizations are used to determine the systemwide least-cost investment and retirement decisions for each location over time, which are the primary outcomes of the model. See the ‘Data’ section below for a description of the data sources used for this analysis, and additional details on the input assumptions and scenarios in ref. 35.

Data

This section summarizes the key data sources and assumptions used for the scenario modelling. Unless otherwise noted, data used are consistent with the assumptions presented in the 2021 Standard Scenarios Report³⁷ and the ReEDS Model Documentation⁴⁶. ReEDS is also publicly available at <https://github.com/NREL/ReEDS-2.0>.

In the following, the data sources for several key parameters are described, and the default assumptions used in the core scenario are highlighted:

- Technology cost and performance assumptions are based on the 2021 Annual Technology Baseline (ATB) of the National Renewable Energy Laboratory⁴⁸. Default assumptions rely on the ‘moderate’ trajectories from the ATB with select scenarios relying on the ‘advanced’ case for offshore wind, other renewable energy and storage technologies. For technologies that are not included in the ATB (for example, direct air capture, electrolyzers and biopower with carbon capture), we adopt the 2022 assumptions of Denholm and co-workers⁴⁹. All scenarios use fossil fuel and uranium prices from the AEO2021 (Annual Energy Outlook 2021) ‘reference’ case of the US Energy Information Administration⁵⁰. By default, nuclear small modular reactors, CCS and carbon dioxide removal are not included as technology options, although select scenarios do include these technologies.
- Transmission costs and loss factors vary between regions, voltage and technology (for example, AC versus DC), and are based on several sources³⁵. Effective distances between regions—for inter-regional transmission and spur lines to connect new renewable-energy plants—are estimated using a least-cost-path algorithm that accounts for costs that vary across terrain, land types and exclusions³⁸. Default assumptions allow AC transmission expansion within each of 12 transmission planning regions approximately based on Order 1000 regions of the Federal Energy Regulatory Commission³⁵. Some sensitivities relax these intraregional constraints on transmission expansion and, in one case, a high-voltage DC macrogrid using voltage-source converter technologies.
- Demand growth is assumed, by default, to follow the trajectory from the Electrification Futures Study (EFS) ‘high electrification’ case^{21,51}, which assumes the growing adoption of end-use electric technologies (for example, plug-in electric vehicles) leading to a compound annual load growth of about 1.9% per

year between 2020 and 2050. Electrification from the EFS trajectory also results in changing load shapes that are due, in large part, to electrified space heating. Sensitivities using a lower-electrification load-growth trajectory, and historical load shapes, are also modelled. For these sensitivities, we use demand growth from the AEO2021 reference case⁵⁰, which projects increases of 0.9% per year in annual demand over the same period. In the high-demand scenarios, the demand growth (of 3.4% per year) and profiles are from ref. 49 and reflect the demand-side changes that are broadly consistent with a net-zero emissions energy system by 2050.

- The renewable resource potential and interconnection costs are based on data from ref. 38 using the Renewable Energy Potential geospatial model^{36,52}. The ‘open access’ offshore-wind-supply curve with few siting exclusions is used by default. A ‘limited access’ supply curve, which requires greater spacing between offshore turbines and various siting exclusions, is also used in a sensitivity analysis. Modelled interconnection costs for offshore wind installations include array costs, marine export cables and onshore interconnection costs. The limited-access siting regimes, which include large setback requirements, are also assumed by default for land-based wind and PV; however, sensitivities using less stringent (‘reference access’) supply curves are also considered for these land-based resources. The different supply curves reflect the varying social acceptance and other potential siting and permitting challenges that are often associated with new clean-energy development. For example, the larger setbacks to structures, roads and other infrastructure under the limited-access siting regime reduce the available resource for development around more densely populated regions³⁶.
- Policies that are modelled include the various national CO₂ emissions constraints described in the main body along with state policies, as of July 2021⁵³. State policies include renewable and clean-energy standards, as well as state offshore wind mandates that are estimated to yield about 19 GW of the new offshore wind capacity by 2030³⁴. The latter include only those mandates that require local utilities to procure offshore wind capacity. Federal renewable-energy tax incentives are also modelled, including an offshore wind investment tax credit of 30% up to 2035. A sensitivity analysis extends the investment tax credit to 2050 for offshore wind energy only. The modelling analysis was conducted before passage of the Inflation Reduction Act of 2022.

Supplementary Tables 1, 2, 3 and 4 describe the key changes from the default assumptions in each scenario. Additional details are also included in ref. 35.

Data availability

The ReEDS model datasets analysed in this article are described in ‘Data’ in the Methods. Source data are provided with this paper.

Code availability

The source code for the ReEDS model is available at <https://github.com/NREL/ReEDS-2.0>. Further information related to the ReEDS code, data and assumptions is available from M.M. (Matthew.Mowers@nrel.gov).

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Author contributions

P.B., T.M. and J.B. conceived and conceptualized the study. M.M., T.M. and P.B. performed the analysis. P.B., T.M. and J.B. wrote the manuscript. M.M. edited the manuscript and provided guidance on the analysis and technical assumptions.

Competing interests

The authors declare no competing interests.

Additional information

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